

OC Core 1.3 / Chemistry Domain Packet

Status Snapshot

- Current state: VALIDATED_ANCHOR_ACTIVE
- Target state: VALIDATED_ANCHOR_ACTIVE
- Scientific class: THEOREM_NATIVE
- Trace status: TRACE_COMPLETE
- Closure verdict: PASS
- Quantitative pass result: PASS
- Next required action: MAINTENANCE_ONLY
- Replay status: PASS_REPLAYABLE

Theorem-to-Observable Map

- K2/K3/K4 organizational thresholds lawfully generate the bounded chemistry packet over thermochemical, spectral, and kinetic observable families.
- The promoted chemistry packet binds source-native thresholds to pinned residual observables without expanding to unrestricted chemical universalization.
- Every promoted chemistry benchmark family is traced through the locked K2/K3/K4 route, pinned reference tables, and held-out replay.

Benchmark Dataset Manifest

- CHEM_BENCH_001: gas_phase_enthalpy_residual | NIST Chemistry WebBook | <https://webbook.nist.gov/chemistry/> | held-out: hold out one compound family from calibration
- CHEM_BENCH_002: spectral_transition_residual | NIST Chemistry WebBook | <https://webbook.nist.gov/chemistry/> | held-out: hold out one spectral family
- CHEM_BENCH_003: kinetic_or_equilibrium_residual | DOE BES / EFRC challenge corpora | <https://science.osti.gov/bes/efrc/Research/Grand-Challenges> | held-out: hold out one reaction-class envelope

Pinned Official Route Protocol

- CHEM_ROUTE_001: NIST (OFFICIAL_CHEMISTRY_REFERENCE) -> <https://webbook.nist.gov/chemistry/> | purpose: Thermochemistry and spectroscopy reference tables
- CHEM_ROUTE_002: U.S. Department of Energy Office of Science (OFFICIAL_CHALLENGE_PROGRAM) -> <https://science.osti.gov/bes/efrc/Research/Grand-Challenges> | purpose: Grand-challenge chemistry benchmark framing

Measurable Outputs

- enthalpy_residual
- spectral_transition_residual
- kinetic_or_equilibrium_residual

Measurement Schema

Map theorem packets to thermochemical, spectral, and kinetic observables against pinned reference tables.

Acceptance Criterion

Declared chemistry benchmark families pass the tolerance envelope on held-out compounds or reactions.

Falsifier

A held-out thermochemical, spectral, or kinetic family outside tolerance collapses the promoted chemistry packet.

Quantitative Acceptance Thresholds

- coverage_ratio_required: 1.0
- critical_residual_sigma_threshold: 2.5
- normalized_error_max_max_sigma: 5.0
- normalized_error_mean_abs_max_sigma: 1.0
- normalized_error_p95_max_sigma: 2.5
- policy_snapshot: {'benchmark_residual_policy': {'critical_residual_sigma_threshold': 2.5, 'schema_id': 'LOGION_BENCHMARK_RESIDUAL_POLICY_v1', 'severe_residual_sigma_threshold': 1.5}, 'prediction_accuracy_policy': {'brier_score_max': 0.08, 'cases_total_required': 30, 'coverage_ratio_required': 1.0, 'critical_failure_f1_min': 0.87, 'critical_failure_precision_min': 0.85, 'critical_failure_recall_min': 0.9, 'expected_calibration_error_max': 0.05, 'schema_id': 'LOGION_PREDICTION_ACCURACY_POLICY_v2'}, 'prediction_sigma_policy': {'cases_total_min': 30, 'normalized_error_max_max_sigma': 5.0, 'normalized_error_mean_abs_max_sigma': 1.0, 'normalized_error_p95_max_sigma': 2.5, 'schema_id': 'LOGION_PREDICTION_SIGMA_POLICY_v1', 'tail_breach_count_max': 0}}
- severe_residual_sigma_threshold: 1.5
- tail_breach_count_max: 0

Replay Harness

- Replay command: `validation/run_all.py --qa-only` (public replay umbrella; domain: CHEMISTRY)
- Evidence bar: HYBRID_ESCALATION
- Protocol id: OC13::EXECUTION::CHEMISTRY::THERMOCHEMISTRY_SPECTRA_KINETICS

Institute-run Escalation

- Current escalation status: NOT_REQUIRED_UNLESS_NEW_OBSERVABLE_FAMILY_ENTERS
- Escalation trigger: OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_CASES_LT_30_OR_RESIDUAL_BREACH_PERSONALITY
- Measurement wave id: INSTITUTE_RUN::CHEMISTRY::WAVE_2A
- Measurement plan: Acquire targeted laboratory or partner measurements for unsupported thermochemical, spectral, or kinetic families before any positive promotion.

Blocking IDs

- None.